



Technology Data Sheet (TDS)

TINMORRY

ASA-CF

Version 1.0

• Basic Introduction

TINMORRY ASA-CF is a carbon fiber reinforced ASA composite. It maintains the excellent strength, temperature resistance, abrasion resistance, durability, outstanding UV resistance, weather resistance and low density of normal ASA. In addition, with the reinforcement of carbon fiber, it has significantly higher strength and stiffness than normal ASA, and efficiently optimizes the warping problem of normal ASA. Besides, it can also endow the prints' surface a fiber feel and look and minimize the layer liness, thereby giving the prints higher dimensional stability, precision and exquisite appearance.

Note:1. Since the ASA-CF filament contains glass fiber, please use a nozzle with a diameter of ≥ 0.4 mm to decrease the risk of nozzle clogging; and please use hard enough nozzles such as those made of hardened steel, and do not use relatively soft nozzles such as those made of brass or stainless steel to slow down the wear of them. 2. It is more recommended to use an enclosed printer for printing, especially for large-sized and high-density models, so as to reduce warping and ensure sufficient layer strength. 3. It will produce a pungent odor during printing, please ensure good ventilation.

• Specifications

Subjects	Data
Diameter	1.75 ± 0.03 mm
Net filament weight	1000 g $\pm$ 0.5%
Length	370 ± 5 m
Spool material	Cardboard (Temperature resistance: 150 °C), or ABS (Temperature resistance: 80 °C)
Spool size	External Diameter: 198 mm; Inner Diameter: 56 mm; Width: 65 mm

• Recommended Printing Settings

Subjects	Data
Drying settings before printing	70 - 80 °C, 6 - 8 h (Blast drying oven, or filament drying box)
Printing temperature and humidity	$\leq 65\text{ °C}$, $\leq 20\%$ RH (Put in an air-tight container and sealed with desiccant, or put it in a filament drying box and set the temp to be about 60 °C)
Storage temperature and humidity	$\leq 30\text{ °C}$, $\leq 20\%$ RH (Sealed with desiccant)
Compatible support material	Itself or support filament for ABS
Compatible printer type	Enclosed-frame (recommended) Open-frame (not recommended)
Compatible nozzle material	Hardened steel, tungsten carbide, etc
Compatible nozzle size	0.4, 0.6 (recommended), 0.8 mm
Compatible plate type	Smooth / Textured PEI Plate or other plate
Plate surface preparation	Glue
Nozzle temperature	260 - 280 °C
Bed temperature	90 - 100 °C (depending on the printer and plate types)
Printing speed*	Max: 300 mm/s
Chamber temperature	$< 65\text{ °C}$

*When printed with Bambu X1C and P1S, and the nozzle temperature is 275 °C, the max printing speed can reach 300 mm/s.

• Properties

The commonly used physical properties, mechanical properties, chemical properties, printing properties and other properties of TINMORRY ASA-CF have been tested and are shown in the tables below.

• Physical Properties

Subjects	Testing Methods	Data
Density	ISO 1183	1.12 g/cm ³
Saturated water absorption rate	25 °C , 55% RH	0.49%
Melt index	260 °C, 2.16 kg	5.7 ± 0.5 g/ 10 min
Glass transition temperature	DSC, 10 °C/min	N / A
Crystallization temperature	DSC, 10 °C/min	N / A
Melting temperature	DSC, 10 °C/min	208 °C
Vicar softening temperature	ISO 306, GB/T 1633	114 °C
Heat deflection temperature 1	ISO 75, 1.82 MPa	95 °C
Heat deflection temperature 2	ISO 75, 0.45 MPa	102 °C

• **Mechanical Properties**

Properties	Testing Methods	Data
Tensile strength (XY)	ISO 527, GB/T 1040	38 ± 3 MPa
Tensile strength (Z)	ISO 527, GB/T 1040	27 ± 3 MPa
Young's modulus (XY)	ISO 527, GB/T 1040	2935 ± 173 MPa
Young's modulus (Z)	ISO 527, GB/T 1040	2278 ± 161 MPa
Breaking elongation rate (XY)	ISO 527, GB/T 1040	$4.2\% \pm 1.1\%$
Breaking elongation rate (Z)	ISO 527, GB/T 1040	$1.3\% \pm 0.7\%$
Bending strength (XY)	ISO 178, GB/T 9341	73 ± 3 MPa
Bending strength (Z)	ISO 178, GB/T 9341	52 ± 3 MPa
Bending modulus (XY)	ISO 178, GB/T 9341	2627 ± 117 MPa
Bending modulus (Z)	ISO 178, GB/T 9341	1981 ± 125 MPa
Impact strength (XY)	ISO 179, GB/T 1043	20.8 ± 1.7 kJ/m ²
Impact strength (Z)	ISO 179, GB/T 1043	9.8 ± 1.3 kJ/m ²

• **Chemical and Other Properties**

Subjects	Data
Component	ASA, carbon fiber
Skin irritation	May cause allergy in some people, and wearing gloves is suggested
Resistance to water	Yes
Resistance to oil and grease	Not resistant to some kinds oil and grease for long-term contact
Resistance to organic solvent	Not resistant to some organic solvents
Resistance to weak acid	Good
Resistance to strong acid	Not resistant
Resistance to weak alkali	Good
Resistance to strong alkali	Not resistant
Flammability	Yes
Odor during printing	Pungent
Odor during combustion	Strongly Pungent

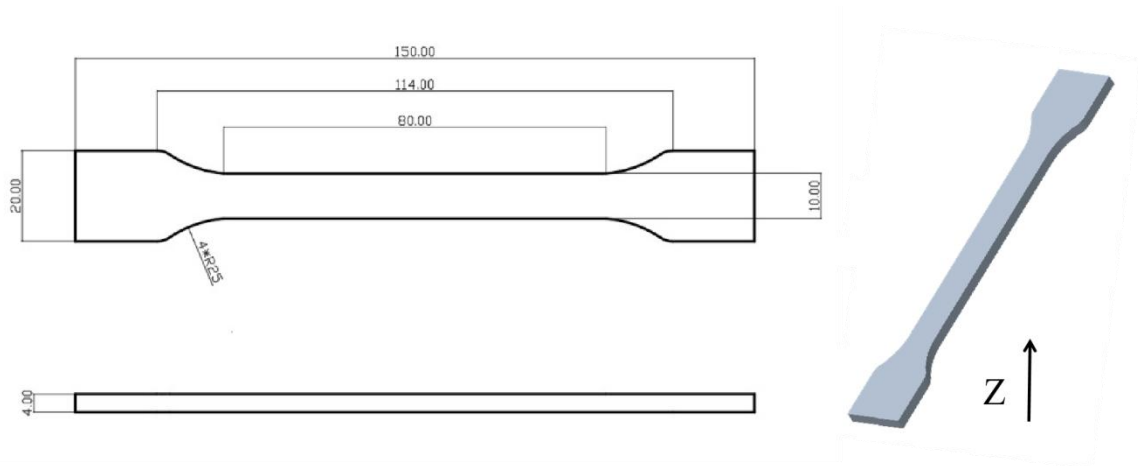
• **Specimen Info**

Subjects	Data
Size	Shown at the following pictures
Nozzle temperature	275 °C
Bed temperature	100 °C
Printing speed	XY: 150 mm/s; Z: 100 mm/s
Infill density	100%
Infill pattern	Concentric

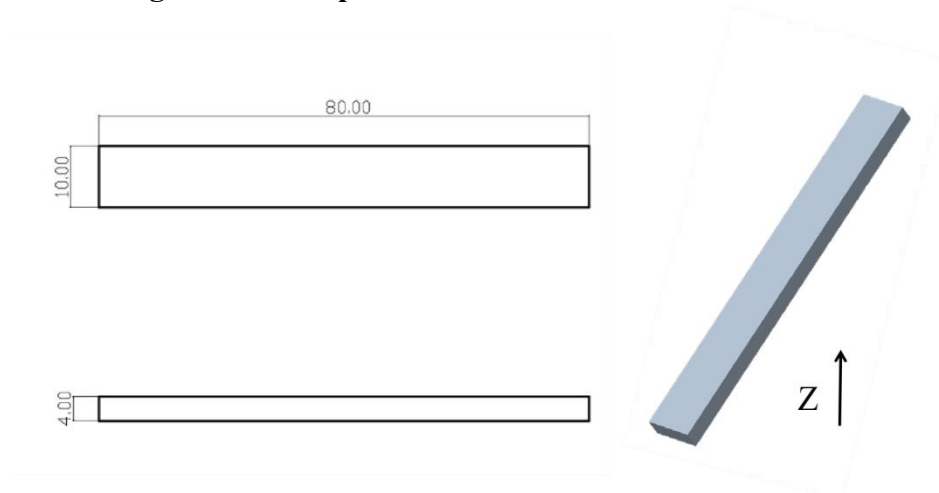
Statements:

1. The test specimens were printed under the key parameters mentioned above. Before testing, all specimens were placed in an indoor environment at about 25 °C for over 12 hours. Please note that when printing, different ambient temperatures, fan speeds and layer times can lead to varying cooling, which can affect the mechanical properties of the specimens, especially those in the Z direction. Additionally, different printers can also result in variations in printing quality.
2. If high printing quality is required, it is recommended to dry every kind of TINMORRY filaments before printing and use an air-tight container and desiccant to protect them from moisture during the printing process (with humidity inside the container less than 20%) to reduce the risk of issues related to moisture absorption in filaments. The recommended drying conditions are shown above. Note that when drying them, avoid placing the spools near the heater to prevent damage to the spools or the filaments due to excessive temperature. Do not use a kitchen oven or microwave oven.
3. After annealing, comprehensive mechanical properties and heat resistance of TINMORRY ASA-CF prints can be slightly increased, while some prints may undergo shrinkage or deformation, so please be aware of this.

1. Tensile Test



2. Bending Test and Impact Test



- **Disclaimer**

The above properties data is obtained by TINMORRY through testing with standard samples, standard methods, and specific equipment, and is for reference and comparison only; if some data are changed, TINMORRY will update this TDS, but may not be able to inform the users, and please forgive us. The actual performance of 3D prints depends not only on the performance of the filaments used, but also on many factors, such as the degree of moisture absorption of the filaments, the printer, environmental conditions, model characteristics, printing parameters, etc.. When using TINMORRY FDM 3D printing filaments, users are responsible for the legality, safety, and performance of the prints. TINMORRY is responsible for the quality of our filaments, but not for any damage or injury that occurs during using them, or the usage scenarios of them.